Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Identify x and y in following reaction:

$2\mathrm{HSO}_4^-(\mathrm{aq}) \xrightarrow{\text{electrolysis}} x \xrightarrow{\text{hydrolysis}} y + 2\mathrm{H}^+(\mathrm{aq}) + \mathrm{H_2O_2(aq)}$

(A) $x = \mathrm{H_2SO_4(aq)}, y = 2\mathrm{HSO}_4^-(\mathrm{aq})$

(B) $x = \mathrm{HO_3SOOSO_3H(aq)}, y = 2\mathrm{HSO}_4^-$ (aq)

(C) $x = \mathrm{HO_3SOOSO_3H(aq)}, \mathrm{y} = \mathrm{H_2SO_4(aq)}$

(D) $x = \mathrm{H_2SO_4(aq)}, \mathrm{y} = \mathrm{HO_3SOOSO_3H(aq)}$

Q.2

Calculate the normality of 10 volume $\mathrm{H_2O_2}$?

Answer upto 2 decimal places only.

(A) 1.78

(B) 1.87

(C) 1.79

(D) 1.97

Q.3

A, 6 volume sample of $\mathrm{H_2O_2}$

(A) will contain $6\%\ \mathrm{V/V}$ of $\mathrm{H_2O_2}$

(B) will contain $6\%\ \mathrm{W/V}$ of $\mathrm{H_2O_2}$

(C) would give 6 volumes of oxygen per unit volume of $\mathrm{H_2O_2}$ sample at STP

(D) would give 6 volumes of oxygen per unit weight of $\mathrm{H_2O_2}$ sample at STP

Q.4

$34\ \mathrm{g}$ of $\mathrm{H_2O_2}$ is present in $1120\ \mathrm{mL}$ of solution. This solution is called

(A) 10 vol solution

(B) 20 vol solution

(C) 34 vol solution

(D) 32 vol solution

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Q.5

The oxidation states of most electronegative element in the products of reaction BaO_2 with dil. H_2SO_4 are

(A) 0 and -1

(B) -1 and -2

(C) -2 and 0

(D) -2 and $+1$

Q.6

A $5.0\ mL$ solution of H_2O_2 liberates $1.27\ g$ of iodine from an acidified KI solution. The percentage strength of H_2O_2 is

(A) 11.2

(B) 5.6

(C) 1.7

(D) 3.4

Q.7

$3.4\ g$ sample of H_2O_2 solution containing $x\%$ H_2O_2 by weight requires xmL of $KMnO_4$ solution for complete oxidation under acidic condition. The normality of $KMnO_4$ solution is

(A) 1 N

(B) 2 N

(C) 3 N

(D) 0.5 N

Q.8

HCl is added to following oxides. Which one would give H_2O_2

(A) MnO_2

(B) PbO_2

(C) BaO

(D) None

Q.9

$100\ mL$ of $0.01M KMnO_4$ oxidises $100\ mL H_2O_2$
in acidic medium. Volume of the same $KMnO_4$ required in alkaline medium to oxidise $100\ mL$ of the same H_2O_2 will be (MnO_4^- changes to Mn^{2+}
in acidic medium and to MnO_2 in alkaline medium)

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(A) $\frac{100}{3}$ mL

(B) $\frac{500}{3}$ mL

(C) $\frac{300}{5}$ mL

(D) None

Q.10

The purity of H_2O_2 in a given sample is 85%. Calculate the weight of impure sample of H_2O_2 which requires 10 mL of $M/5 KMnO_4$ solution in titration in acidic medium

(A) 2 g

(B) 0.2 g

(C) 0.17 g

(D) 0.15 g

Q.11

Barium peroxide reacts with phosphoric acid to produce barium phosphate along with

(A) water

(B) hydrogen peroxide

(C) dioxygen

(D) phosphine

Q.12

The normality and volume strength of a solution made by mixing 1.0 L each of 5.6 volume and 11.2 volume H_2O_2 solution are :

(A) 1 N, 5.6 vol

(B) 1.5 N.5.6vol

(C) 1.5 N, 8.4vol

(D) 1 N, 8.4 vol

Q.13

The amount of H_2O_2 present in 1 litre of 1.5 NH_2O_2 solution, is :

(A) 25.5 g

(B) 3.0 g

(C) 8.0 g

(D) 2.5 g

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Q.14

From the following statements regarding H_2O_2, choose the incorrect statement :

(A) It has to be stored in plastic or wax lined glass bottles in dark

(B) It has to be kept away from dust

(C) It can act only as an oxidizing agent

(D) It decomposes on exposure to light

Q.15

Calculate the amount of H_2O_2 present in 1 litre of $1.5NH_2O_2$ solution.

(A) 25.5

(B) 25.6

(C) 25.7

(D) 25.4

Q.16

What will be the mass of oxygen liberated by decomposition of 200 mL hydrogen peroxide solution with a strength of $34\ g$
per litre?

(A) 2.3

(B) 3.2

(C) 3.5

(D) 5.3

Q.17

$H_2O_2 \rightarrow 2H^+ + O_2 + 2e^-; E^\circ = -0.68\ V$. This
equation represents which of the following behaviour of H_2O_2?

(A) Reducing

(B) Oxidising

(C) Acidic

(D) Catalytic

Q.18

H_2O_2 is always stored in black bottles because

(A) It is highly unstable

(B) Its enthalpy of decomposition is high

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(C) It undergoes auto-oxidation on prolonged standing

(D) None of these

Q.19

Which of the following is wrong about H_2O_2? It is used

(A) as an aerating agent in production of sponge rubber

(B) as an antichlor

(C) for restoring white colour of blackened lead painting

(D) None of these

Q.20

What is the false about H_2O_2 ?

(A) Acts both as oxidising and reducing agent.

(B) It is an antiseptic and germicide for working wounds, teeth and ears under the name perhydrol.

(C) H_2O_2 can be distilled under reduce pressure.

(D) Two OH bonds of H_2O_2 lie in the same plane.

Answers & Solutions

Q.1 Answer:

$x = HO_3SOOSO_3H(aq), y = 2HSO_4^-$ (aq)

Solution:

$$2HSO_4^-(aq) \xrightarrow{\text{electrolysis}} HO_3SOOSO_3H(aq)$$

$$\text{Hydrolysis} \longrightarrow 2HSO_4^-(aq) + 2H^+(aq) + H_2O_2(aq)$$

Q.2 Answer:

1.79

Solution:

Normality of 10 V **of** H_2O_2

$\frac{68\times10}{22.4} = 17 \times N \quad \therefore N = 1.78$

$N = 1.7857$ **which when rounded off to 2 decimal places will be** 1.79**.**

Hence (3) answer.

Q.3 Answer:

would give 6 volumes of oxygen per unit volume of H_2O_2 sample at STP

Solution:

6 volume H_2O_2 **would give 6 volumes of** O_2 **per unit volume of** H_2O_2

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Q.4 Answer:

10 vol solution

Solution:

$\mathrm{N}=\frac{\mathrm{W}_2 \times 1000}{\mathrm{E} \cdot \mathrm{w}_2 \times \mathrm{VmL}}$
$=\frac{34 \times 1000}{17 \times 1120}=\frac{200}{12}$
N **of** $\mathrm{H_2O_2}=5.6$ **vol strength**
$\frac{200}{12}$ N **of** $\mathrm{H_2O_2}=5.6 \times \frac{200}{12}=10$
vol strength

Q.5 Answer:

-1 and -2

Solution:

$\mathrm{BaO_2}+\mathrm{H_2SO_4} \rightarrow \mathrm{BaSO_4}+\mathrm{H_2O_2}$.
oxygen has common O.S. as -2 **and in peroxides as** -1.

Q.6 Answer:

3.4

Solution:

$\mathrm{H_2O_2} \equiv \mathrm{KI} \equiv \mathrm{I_2}$
mEq **of** $\mathrm{H_2O_2}=5 \times \mathrm{N}$
mEq **of** $\mathrm{I_2}=\frac{\text{Weight}}{\mathrm{E.w}} \times 1000$
$\frac{1.27}{127} \times 1000=10 \mathrm{mEq}$
$\mathrm{I_2}=\mathrm{M.w}=254$
$\mathbf{E.W}=\frac{M.w}{2}=\frac{254}{2}=127$
$\therefore \mathrm{mEq}$ **of** $\mathrm{H_2O_2}=\mathrm{mEq}$ **of** $\mathrm{I_2}$
$5 \times \mathrm{N}=10 \quad \therefore \mathrm{N}=2$
$1 \mathrm{~N}$ **of** $\mathrm{H_2O_2}=1.7 \%$ **of** $\mathrm{H_2O_2}$
$2 \mathrm{~N}$ **of** $\mathrm{H_2O_2}=3.4 \% \mathrm{H_2O_2}$

Q.7 Answer:

2 N

Solution:

$100 \mathrm{~g}$ **of** $\mathrm{H_2O_2}$ **sample solution contains** $=\mathrm{x} \mathrm{~g}$ **of** $\mathrm{H_2O_2}$
$3.4 \mathrm{~g}$ **of** $\mathrm{H_2O_2}$ **sample solution contains** $=\frac{x}{100} \times 3.4$
Weight of $\mathrm{H_2O_2}=\frac{3.4 \mathrm{x}}{100}$
Eq. of $\mathrm{H_2O_2}=\frac{3.4 \mathrm{x}}{100} \times \frac{1}{17}$
mEq **of** $\mathrm{H_2O_2}$
$=\frac{3.4 \mathrm{x}}{100 \times 17} \times 1000=\frac{34 \mathrm{x}}{17}=2 \mathrm{x}$
mEq **of** $\mathrm{KMnO_4}=\mathrm{x} \times \mathrm{N}$

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$x \times N = 2x$
$N = 2$

Q.8 Answer:

None

Solution:

MnO_2, PbO_2 **and BaO will not give** H_2O_2
with HCl. MnO_2 **and** PbO_2 **will give** Cl_2 **and** BaO **will react with** HCl **to give** $BaCl_2$ **and water.**

Q.9 Answer:

$\frac{500}{3}$ mL

Solution:

$MnO_4^- + 5e^- \rightarrow Mn^{2+}$
$MnO_4^- + 3e^- \rightarrow MnO_2$
$100\ mL H_2O_2 \equiv 100 \times 5\ N$
$MnO_4^- \equiv V \times 3\ N MnO_2$
$N = \frac{500}{3}$ mL

Q.10 Answer:

0.2 g

Solution:

mEq **of** $MnO_4^- \equiv$ mEq **of** H_2O_2
$10 \times \frac{M}{5} \times 5 \equiv$ mEq **of** H_2O_2
$\Rightarrow 10 mEq \equiv$ mEq **of** H_2O_2
Weight of $H_2O_2 = 10 \times 10^{-3} \times \frac{34}{2} \times 0.17$
$= 0.2$ g
Weight of impure H_2O_2
$= \frac{100}{85} \times 0.17 = 0.2$ g

Q.11 Answer:

hydrogen peroxide

Solution:

$3BaO_2 + 2H_3PO_4 \rightarrow Ba_3(PO_4)_2 + 3H_2O_2$

Barium Phosphoric Barium Hydrogen peroxide
acid phosphate peroxide

Q.12 Answer:

1.5 N, 8.4vol

Solution:

$1\ N = 5.6 vol$
$\Rightarrow$ **Normalities of two solutions are** $1\ N$ **and** $2\ N$

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Normality of mixture
$= \frac{1\times1+1\times2}{1+1} = \frac{3}{2} = 1.5\ \text{N}$
Volume strength
$= \frac{3}{2} \times 5.6 = \frac{16.8}{2} = 8.4\text{vol}$

Q.13 Answer:

$25.5\ \text{g}$

Solution:

Molecular weight of $H_2O_2 = 34$ **Equivalent weight of** $H_2O_2 = 17 \therefore 1\ \text{L}$ **of** $1\ NH_2O_2 \text{has} = 17\ \text{g}$ **of** H_2O
$\therefore 1\ \text{L}$ **of** $1.5\ NH_2O_2$ **has**
$= 1.5 \times 17 = 25.5\ \text{g}$ **of** H_2O_2

Q.14 Answer:

It can act only as an oxidizing agent

Solution:

H_2O_2 **has oxidizing and reducing properties both.**

Q.15 Answer:

25.5

Solution:

Molecular weight of $H_2O_2 = 34$ **Equivalent weight of** $H_2O_2 = 17 \because 1\ \text{L}$ **of** $1\ NH_2O_2 = 17\ \text{g}$ **of** H_2O_2
$\therefore 1\ \text{L}$ **of** $1.5NH_2O_2 = 1.5 \times 17 = 25.5\ \text{g}$ **of** H_2O_2

Q.16 Answer:

3.2

Solution:

Strength of the solution $= 34\ \text{g/L}$
$1\ \text{L}$ **of the solution contains** $34\ \text{g}$ **of** H_2O_2
$200\ \text{mL}$ **of the solution contains** $\frac{34}{1000} \times 200 = 6.8\ \text{g}$
of H_2O_2
$2H_2O_2 \rightarrow 2H_2O + O_2$
$2 \times 34\ \text{g} \quad 32\ \text{g}$
$68\ \text{g}$ **of** H_2O_2 **gives** $32\ \text{g}$ **of** O_2
$6.8\ \text{g}$ **of** H_2O_2 **gives** $= \frac{32}{68} \times 6.8 = 3.2\ \text{g}$ **of** O_2

Q.17 Answer:

Reducing

Solution:

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As H_2O_2 is loosing electrons, so it is acting as reducing agent.

Q.18 Answer:

It undergoes auto-oxidation on prolonged standing

Solution:

H_2O_2 is unstable liquid and decomposes into water and oxygen either on standing or on heating

Q.19 Answer:

None of these

Solution:

H_2O_2 show all these properties.

Q.20 Answer:

Two OH bonds of H_2O_2 lie in the same plane.

Solution:

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